

NATHAN KIMUTAI LOTOBO

FRONTEND • SOFTWARE ENGINEER • AI ENGINEER

Kenya | 0753 390 828 | nathankimutai48@gmail.com | github.com/kim214 | linkedin.com/in/nathan-kimutai-163a9036b

PROFESSIONAL SUMMARY

Software/IT engineer with experience in software development, systems administration, technology leadership, application delivery, and security-focused systems management. Experienced in designing and maintaining software solutions, protecting organizational systems against unauthorized access, coordinating technology projects, and working across development and infrastructure environments. Strong foundation in Python, JavaScript, Java, Kotlin, SQL, Git/GitHub, Node.js, React Native, databases, APIs, IoT, and AI. Seeking to apply this engineering mindset to application security, secure software development, vulnerability reduction, and developer-focused security enablement.

CORE COMPETENCIES

Application & Systems Security • Secure Software Development • Access Control & Authentication • Security Risk Awareness • Vulnerability Identification & Remediation • Software Architecture • API & Web Security Fundamentals • Git/GitHub Workflows • DevSecOps Fundamentals • CI/CD Concepts • Database Security • Project & Technical Leadership • Security Documentation • AI/IoT Systems • Cross-functional Collaboration

PROFESSIONAL EXPERIENCE

Head of IT — African Union Students Platform | 01/2025 – Present

- Lead the organization's IT function, overseeing the design, implementation, maintenance, and security of digital systems and infrastructure.
- Establish and support security controls designed to protect applications, systems, and organizational data from unauthorized access and security threats.
- Provide technical leadership on software solutions, system reliability, access management, security practices, and continuous technology improvement.
- Lead technology projects from planning through implementation, coordinating stakeholders and technical work to deliver secure, reliable solutions aligned with organizational objectives.
- Translate technical and security requirements into practical guidance for users and project stakeholders.

IT & Systems / Lead Software and Systems Engineer — Uniglobe Access | 04/2020 – 07/2026

- Maintain and support organizational software systems, focusing on reliability, availability, security, and effective day-to-day operation.
- Design and improve software solutions while implementing measures that help prevent unauthorized access and protect organizational systems.
- Support system security, access controls, and operational practices to reduce exposure to avoidable security risks.
- Coordinate and manage projects, tracking implementation activities and stakeholders to ensure efficient, high-quality delivery.
- Work across technical and operational requirements to align project execution with organizational objectives.

Freelance Software & Design Developer | Self-Employed / Remote | 2022 – 2025

- Develop websites and applications tailored to client requirements using modern software development technologies.
- Work across front-end, back-end, databases, APIs, and version-control workflows to deliver functional software solutions.
- Translate requirements into practical technical solutions while working to deadlines and maintaining solution quality.
- Collaborate directly with clients to understand requirements, troubleshoot issues, and deliver iterative improvements.

PROJECTS

RecruitIQ — AI-Powered Recruitment Platform

- Built a platform that streamlines hiring by automatically scoring and ranking applicants against role requirements, cutting down manual screening time.
- Generates detailed, easy-to-read candidate reports and conducts AI-driven interviews to give recruiters a consistent, data-informed view of every applicant.

- Pitched for the Execute Africa AI Challenge, with academic documentation covering system architecture and design (UML, DFD, and conceptual framework diagrams).

GreenTech — Automated Smart Greenhouse System

- Designed and built an end-to-end smart-agriculture solution that gives farmers real-time visibility and control over greenhouse conditions from a mobile app.
- Automates irrigation, ventilation, and pest control based on live environmental data, reducing manual monitoring and helping protect crop yields.
- Led the project as Project Lead, taking it from concept through implementation, including engagement with the ICT Authority Kenya on platform registration.
- Recognized with a Bronze Medal at the China International College Students' Innovation Competition (2025).

MahindAI — Maize Health Assistant

- Developed a mobile and web app that helps maize farmers diagnose crop diseases from a simple leaf photo, delivering an instant diagnosis with severity and treatment guidance.
- Built in an AI chat assistant so farmers can ask questions about diseases, pests, and farm management, backed by a full disease encyclopedia and seasonal planting calendar.
- Designed for real-world East African farming contexts, with a bilingual English/Swahili interface to widen accessibility.

AI First Aid Assistant (AIFAA)

- Developed an AI-enabled mobile app that guides users through first-aid situations step by step in real time, prioritizing clarity and responsible, safe use of automated guidance.
- Built with offline-first functionality and multilingual support so it stays usable in areas with limited or unreliable connectivity.
- Selected to participate in the AfyaVentures bootcamp as part of a three-person team.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, JavaScript, C, SQL, HTML/CSS

Frameworks & Development: React Native, Node.js, Android Studio, Prisma, REST/API development

Security & Engineering: Access control, authentication concepts, secure software practices, vulnerability awareness, Git-based workflows, security-focused system maintenance

Databases & Infrastructure: MySQL, PostgreSQL, Git/GitHub, Docker, WSL, MQTT

AI & Data: TensorFlow, scikit-learn, Pandas, Matplotlib

IoT: ESP32, Arduino, sensors, actuators, automation systems

EDUCATION

Bachelor of Science in Computer Science — Kabarak University, Nakuru, Kenya | 2023 – 2027 (Expected)

Kenya Certificate of Secondary Education (KCSE) — Sacho High School | 2019 – 2023

CERTIFICATIONS, LEADERSHIP & RECOGNITION

- Treasurer, Kabarak University Innovation Club — 2024–Present
- Project Lead, Green Tech Automated AI/IoT Greenhouse System — 2024–Present
- Bronze Medalist, China International College Students' Innovation Competition — 2025
- Participant, Central Rift Innovation Week — Kabarak University, 2025
- Participant, Kenya National Research Festival — Egerton University, 2025
- Effectively Engage Within the UNFCCC Climate Change Negotiation Process — FAO/UNFCCC, 2024
- Climate Change Negotiations for Agriculture Stakeholders — FAO/UNFCCC, 2024
- Geothermal Energy Training — Geothermal Development Company, 2025